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A scoping review of interventions, services, and models of care to improve access to cancer treatment for adults who are socially disadvantaged

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ABSTRACT

Background: Timely and equitable access to cancer treatment are important markers of the quality and accessibility of cancer care, yet socially disadvantaged people experience a range of barriers to receiving cancer treatment, significantly impacting cancer outcomes, including survival and quality of life. However, interventions designed to improve treatment access for these populations have not been comprehensively mapped. This scoping review aimed to identify and summarize interventions, services, and models of care intended to improve access to cancer treatment among socially disadvantaged groups.

Methods: We conducted a scoping review to identify and summarize interventions designed to improve access to cancer treatment for people experiencing varying forms of social disadvantage. MEDLINE, EMBASE, and CINAHL were searched for relevant studies published before 2025. Data on study characteristics, intervention types, targeted barriers, and outcomes related to treatment access were extracted and synthesized descriptively. A formal risk-of-bias assessment was not conducted, consistent with scoping review methodology.

Findings: A total of 57 studies met the inclusion criteria, with nearly half published since 2020. Most studies were conducted in the United States (56) and used observational designs. Interventions addressed various treatment modalities, most commonly surgical treatment (37%), followed by radiotherapy and systemic therapy. Policy-focused interventions, particularly Medicaid expansion, were most frequently evaluated, followed by patient navigation programs. Education-based interventions were relatively few (7%). Institutional and system-level interventions showed more consistent improvements in treatment access. The barriers most commonly targeted included insurance coverage, financial constraints, and transportation.

Conclusions: Most interventions focused on financial or insurance-related disadvantage, highlighting gaps in evidence-informed strategies addressing other forms of social disadvantage. The predominance of U.S.-based studies also indicates a need for evidence on effective interventions in other health system contexts.

Introduction

> *"Cancer is a disease of difference not only at the micro or molecular level, but also at the macro or societal level"* (Wild, 2019, p. 1).

Social disadvantage is both a cause and a consequence of unequal access to resources and social determinants of health (e.g., income, education, housing, employment). Socially disadvantaged populations also experience unequal access to power and social capital resulting from

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systems of oppression including colonialism, racism, sexism, and ableism. Broader social structures play an additional role in creating and perpetuating social disadvantage, including ideological, economic, political and historical contexts; the interconnected nature of social and structural determinants of health is explicated in the World Health Organization's Conceptual Framework for Action on the Social Determinants of Health (Solar and Irwin, 2010). These landmark reports drew significant attention to the influence of non-biological factors on health equity and health outcomes; in the past decade, specific attention to cancer-related inequities has followed. Specifically, the relationship between socioeconomically disadvantaged populations and disparities in cancer outcomes has been established through the work of international and national reports (American Association of Cancer Research, 2022; Vaccarella et al., 2023) and the importance of health equity as a policy goal strengthened through the work of national and international policy statements and cancer plans (Australian Government, Cancer Australia, 2026; Canadian Partnership Against Cancer, 2019; European Union, 2022; Patel et al., 2020). Although most often understood as a biological disease, the case continues to strengthen for also understanding cancer as having strong links to social and structural determinants of health (Krieger, 2019).

Timely access to guideline-recommended cancer treatment is a known indicator of the quality and accessibility of a cancer care system, and one of several factors impacting cancer outcomes (Canadian Partnership Against Cancer, 2014). However, populations experiencing various forms of social disadvantage experience inequities in access to cancer treatment, including delays in treatment initiation, differences in receipt of treatment, and treatment that does not align with standards of care. For example, evidence suggests that socially disadvantaged adults are less likely to receive consultation by an oncologist following a cancer diagnosis (Davis et al., 2020), less likely to receive cancer treatment (AACR, 2022; Canadian Partnership Against Cancer, 2020; Lejeune et al., 2010; Shah and Chan, 2021), or may be offered treatment that is of lower quality (AACR, 2022; Lineback et al., 2016; Zavala et al., 2021). These inequities have significant impacts on cancer outcomes and quality of life, including an ever-growing body of evidence demonstrating disparities in cancer-related mortality and survival globally (AACR, 2022; Fan et al., 2022; Forrest et al., 2013; Vaccarella et al., 2023, 2019).

Contributing to these inequities are a host of intersecting barriers to accessing cancer treatment among people experiencing social disadvantage. Unequal access to social determinants of health creates unmet social needs, including challenges related to housing and food insecurity; meeting daily needs for survival can delay or limit access to cancer treatment (Bourgeois et al., 2024, 2023; Fan et al., 2022; Horrill et al., 2019). Although unmet social needs are themselves a barrier, our research also suggests that the lack of assessment of social needs also contributes to access issues (Bourgeois et al., 2023; Horrill et al., 2025a). Lower levels of health literacy also disproportionately impact socioeconomically disadvantaged populations, which have particular impacts on cancer treatment, including patients who have a limited understanding of treatment options and treatment regimens, resulting in delayed treatment, missed treatments, and/or decisions to decline treatment altogether (Bourgeois et al., 2023; Lawrie et al., 2020; Lineback et al., 2016). The complexity and fragmented nature of many health systems, often intersecting with unmet social needs and lower health literacy, further creates barriers across the cancer care continuum (Black et al., 2021; Bourgeois et al., 2024). Moreover, the delivery of cancer treatment services, which tends to be organized around standardized, protocol-driven care to maximize efficacy and safety of treatment and prolong survival, creates particular challenges for people experiencing social disadvantage, who may not have the resources and social supports to manage the rigorous demands of cancer treatment (Bourgeois et al., 2024; Horrill et al., 2025a). Cancer treatment services are also known to be concentrated in urban areas, making geographical location of residence, and rurality in particular, a significant barrier that is distinct from, but often compounds other forms of social disadvantage (Bourgeois et al., 2024). Many socially disadvantaged people will have experiences of trauma, stigma, discrimination, and/or racism that intersect to create additional barriers or compound existing barriers to cancer treatment (Bourgeois et al., 2024; Horrill et al., 2022, 2019).

Despite these challenges, interventions, services, and models of care have been developed with the explicit aim of improving access to care for socially disadvantaged populations across the cancer care continuum. Most notably, patient navigation as an intervention was first developed and implemented in the 1990's by American surgical oncologist Harold Freeman to improve access to breast cancer screening and timely follow-up for socioeconomically disadvantaged women in Harlem, New York by providing personalized support and guidance to address barriers to care (Freeman and Rodriguez, 2011). Since then, patient navigation has expanded to include both professional (e.g., oncology nurses) and lay (e.g., peer) models of navigation across the cancer continuum, with evidence suggesting patient navigation is effective in addressing multiple barriers to care (Chan et al., 2023; Chen et al., 2024). Although patient navigation shows promise as an intervention aimed at improving access to cancer treatment for socially disadvantaged populations, additional services, models of care and/or interventions targeted at the individual, organizational, systems and policy levels are needed to tackle persistent inequitable cancer health outcomes, while also addressing other aspects of the quintuple aim for healthcare improvement such as enhancing care experiences (Nundy et al., 2022).

The purpose of this scoping review was to examine interventions, services and models of care (herein "interventions") specifically designed to improve access to or receipt of cancer treatment among people who are socially disadvantaged. A scoping review was selected as the most appropriate methodology to develop a comprehensive overview of the evidence base, identify gaps in evidence, and inform intervention development and/or adaptation in this field. Importantly, we recognize that equity-promoting interventions in cancer care are an emerging field, and the evidence may vary in terms of quality, methodological approach, and setting; a scoping review allows for inclusion of a range of evidence (Peters et al., 2021a, 2021b). Preliminary literature searches conducted by the academic librarian (JL) at the time of protocol development indicated there were no other planned or ongoing reviews on this topic. Existing reviews have primarily synthesized literature on disparities in access to cancer services (Bourgeois et al., 2024; Grant et al., 2024; Horrill et al., 2019), focused on single intervention types (Chan et al., 2023), or have been limited by country and timeframe (Santos Salas et al., 2024).

Methods

Our scoping review approach is based on the framework proposed by Arksey and O'Malley (2005) with enhancements by Levac et al. (2010), and guidance from Peters et al. (2021a, 2021b). Our review was planned and implemented in six iterative stages, as outlined below, and reported based on the PRISMA Extension for Scoping Reviews (PRISMA-ScR) (Moher et al., 2015; Tricco et al., 2018). The protocol was previously published (Horrill et al., 2024) and registered in Open Science Framework (https://doi.org/10.17605/OSF.IO/83ANZ).

This review was grounded in critical, intersectional, and social justice perspectives. From a social justice perspective, health and healthcare access are understood as basic human rights, and as shaped by socioeconomic, political and historical contexts (Reimer Kirkham and Browne, 2006). Health inequities, and by extension, inequities in cancer outcomes and access to care are arguably issues of social justice which require intervention (Horrill et al., 2025b; Reimer Kirkham and Browne, 2006; Varcoe et al., 2014). Critical theoretical perspectives aim to develop practical knowledge to redress a particular injustice or inequity (Kincheloe and McLaren, 2005). Underpinned by social justice aims, critical theoretical perspectives focus our attention on how multifaceted

social, political, and power dynamics impact the design and delivery of healthcare services (Strega, 2015). These perspectives recognizing the complex interplay of macro, meso and micro factors, including how health systems operate and either facilitate or limit healthcare access at these levels. Intersectionality theory, originating in the early 19th century among Black American women activities, and advanced through the writings of Crenshaw, Collins and others, assumes that forms of social disadvantage and oppression are interdependent and compounding rather than additive or acting in parallel (Crenshaw, 1991; Patricia Hill Collins, 2000). Through an intersectional lens, aspects of identity and oppression intersect in complex and co-constitutive ways to create and perpetuate cancer health inequities. These perspectives have informed our attention to how social disadvantage is conceptualized and operationalized in research, and our focus on actively addressing inequities for socially disadvantaged populations.

Identifying the research question

Based on the growing evidence pointing toward inequities in access to cancer treatment among socially disadvantaged populations, and the emerging prioritization of equity in cancer care, our scoping review aimed to answer the following question: "What interventions, services, or models of care have been developed to improve access to or receipt of cancer treatment for adults who are socially disadvantaged?" Our specific objectives were two-fold:

1. To identify the extent and summarize the nature of interventions, services, or models of care that have been developed and implemented with the explicit aim of improving access to cancer treatment for adults who are socially disadvantaged, including how they are designed, organized and delivered, and the types of strategies used to improve access to cancer treatment;
2. To summarize how these interventions, services, or models of care have been evaluated, and key findings from evaluations.

Concepts central to our research question included 'cancer treatment' and 'socially disadvantaged populations', and are defined in Table 1.

Identifying relevant studies

A comprehensive search strategy was designed by an academic librarian (JL) with extensive expertise in research and literature searching with structurally marginalized populations. Primary databases searched included OVID Medline, EMBASE, and CINAHL (using EBSCOhost); ProQuest was used to search for relevant dissertations. These searches were conducted in August 2023 Google Advanced was used to identify relevant online grey literature reports. Citation searching (Hirt et al., 2023) was conducted between June and December 2024 using several methods: a) screening reference lists of relevant systematic reviews identified in the database searches; b) backward searching by screening reference lists of articles included for data extraction; and c) forward searching by screening references that cited the article included using Scopus. No date limits were set; however, retrieval of grey literature and dissertations was limited to those accessible online. Search results were merged and de-duplicated using Covidence software. Database search strategies are included in Supplemental File 1.

Table 1
Key concepts as defined in review protocol.

Concept	Definition
Cancer treatment	Primary therapies used to stop, shrink, or slow down the progression of cancer, including surgery, radiotherapy, systemic therapy (including chemotherapy, targeted therapy, immunotherapy and hormone therapy), stem cell and bone marrow transplant (Tortorice, 2018.).
Social disadvantage	Relative position in the social order, influenced and stratified by access to the social determinants of health (e.g., income, employment, housing) as well as aspects of one's identity, including skin color, national origin, religion, age, gender, gender identity, sexual orientation, disability, or mental health status (Braveman, 2014; Solar and Irwin, 2010). Root causes of social disadvantage are intersecting and multiplicative (Dhamoon and Hankivsky, 2011).
Healthcare access	A fluid and dynamic process, which includes the opportunity and ability to obtain and/or use healthcare services, mediated by the availability, affordability, quality, timeliness, and ability to reach services; the degree to which people attempting to obtain services are deemed as 'eligible' to use or receive those services; and the degree to which services are perceived by users as socially safe and acceptable (Dixon-Woods et al., 2006; Horrill et al., 2018; Levesque et al., 2013).

Study selection

Study selection proceeded in an iterative manner, with inclusion and exclusion criteria evolving as we reviewed and discussed articles for

Table 2
Inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
Language of Publication	English	All others
Population	Population experiencing social disadvantage along one or more axis, including (but not limited to): • Economic status (low income, poverty) • Educational attainment • Unstably housed • Member of a racialized group • Geography (rural, remote) • Marginalized by way of: religion, national origin, immigrant status, gender, gender identity, sexual orientation, disability, mental health status, or substance use Adult (18 +)	Population not experiencing social disadvantage Pediatric/non-adult population
Focus of Publication	• Describes a model of care, service, or intervention (including policy intervention) developed and implemented with the explicit aim of *improving access to* (defined above) or *receipt of* cancer treatment, including treatment specific cancer clinical trials • Cancer treatment: treatments to stop, shrink or slow the progression of cancer, including surgery, radiotherapy, systemic therapy (chemotherapy, targeted therapy, immunotherapy, and/or hormone therapy), stem cell transplant, or bone marrow transplant. • The described service or model of care has been evaluated and evaluation data is available	
Type of publication	• Peer-reviewed primary research studies (including observational studies, randomized controlled trials, quasi-experimental studies and other non-randomized trials, systematic reviews, qualitative studies) • Peer-reviewed quality improvement reports	Non-peer reviewed research, case reports, discussion papers, opinion papers, commentaries, conference proceedings, slide presentations, news stories

inclusion. Inclusion and exclusion criteria are detailed in Table 2. We used a 3-step process for study selection. First, a single reviewer (PO) completed screening of titles of all articles identified through electronic searches, excluding any articles that were clearly irrelevant (for example, articles clearly not related to cancer or health services). Second, two reviewers independently screened abstracts of remaining articles against the inclusion criteria. Reviewers (AB, MS, PO, JC) met weekly with the PI (TH), and discrepancies were resolved by consensus. Finally, two reviewers independently reviewed each full text of articles included in Step 2 for eligibility, noting the reason for exclusion, where applicable (AB, MS, NB, JC). As in Step 2, reviewers met weekly with the PI to resolve discrepancies by consensus. Study selection was managed using Covidence software (www.covidence.org) and is reported as per the PRISMA Extension for Scoping Reviews (Fig. 1).

Data charting

Two reviewers (MS, NB) extracted data from the included full-text articles using a standardized extraction form. The data extraction form was developed iteratively, to meet the aims of the review: we identified key elements for data charting and drafted an initial data extraction form prior to data extraction, pilot tested it with five articles, subsequently revised the form and pilot tested it with a second set of five articles prior to finalizing the form (see Supplemental File 2). To ensure a consistent and robust approach to data charting, reviewers met with the PI bi-weekly throughout this process. Data extraction and consensus was managed using Covidence software.

Collating, summarizing and reporting the results

In the final stage of our review, we engaged in an iterative process of data analysis and reporting of results to address the aims of the review and identify key gaps in the literature and implications of our findings. Analysis of data was led by the first author, who brought preliminary findings and implications to the interdisciplinary research team for discussion and critique. We also engaged members of our Community Advisory Board to discuss preliminary findings and implications. Feedback was used to inform further analysis, identify implications, and

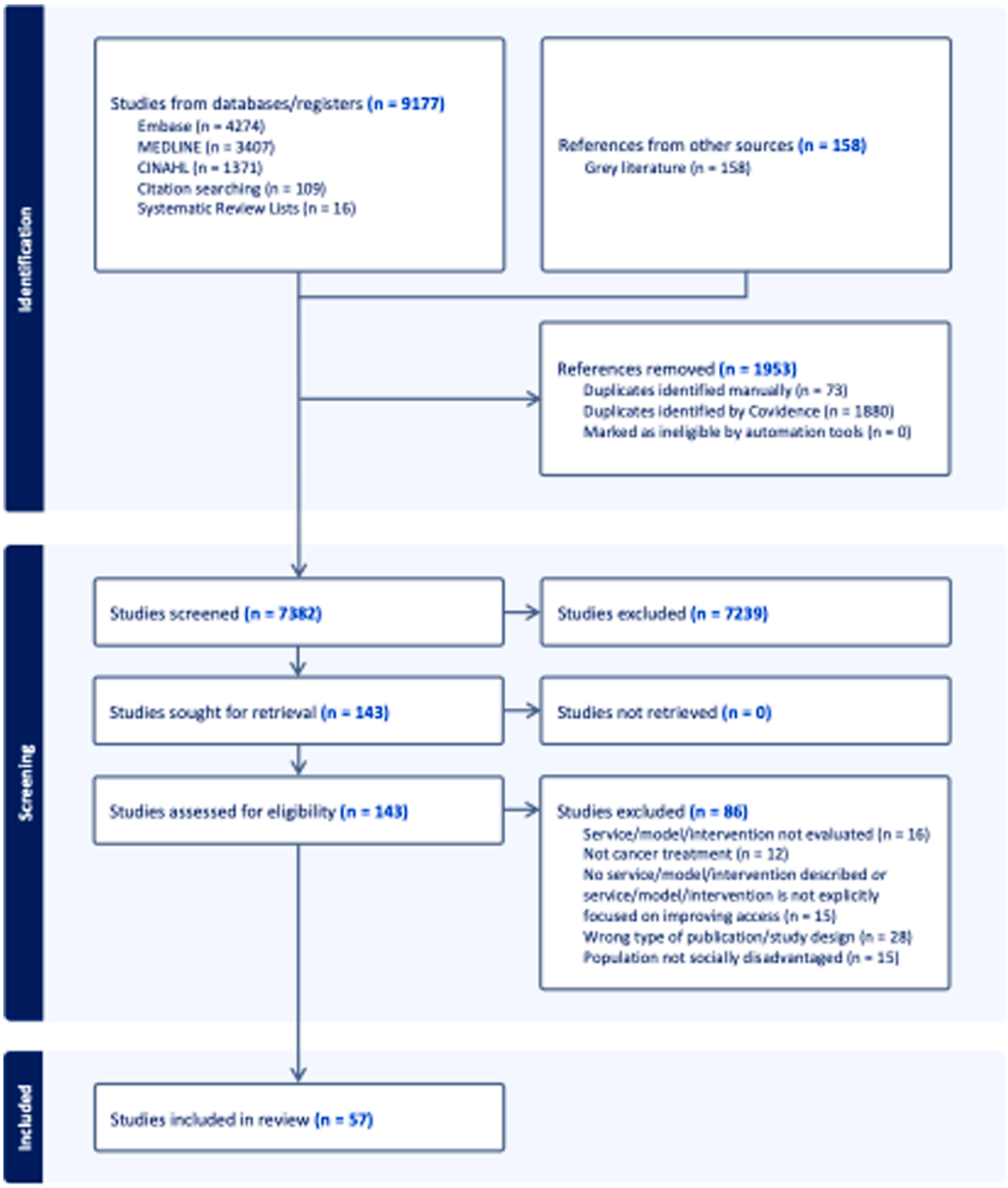


Fig. 1. Study Selection – PRISMA Flow Diagram.

articulate future directions. The results of this review are reported in a narrative summary of findings.

Findings

A total of 57 studies were included in our analysis; all were peer-reviewed published papers, and nearly all were studies conducted in the United States (n = 56) with a single study conducted in Australia. Although we did not limit our inclusion by year, the earliest included study was published in 2003, with a clear trend to increasing publications of relevant studies seen between 2008 and 2024 (Fig. 2).

Study characteristics are shown in Table 3. More than half of included studies reported on interventions focused on only a single type of cancer (n = 31), while one third focused on multiple types of cancer (n = 20); interventions most commonly focused on gastrointestinal (GI; n = 28), breast (n = 27), genitourinary (GU; n = 19) and lung (n = 15) cancers. The number of studies reporting on interventions focused on a single treatment type was similar to the number focused on multiple types of treatment (n = 26, n = 31 respectively). Most studies (n = 40) specified a singular primary social disadvantage variable of interest; the remaining 17 studies specified multiple social disadvantage variables without identifying a singular primary variable, suggesting an intersectional approach to studying social disadvantage. Of the 40 studies that identified a primary social disadvantage variable, the most common variables were Medicaid enrolled/qualified (n = 18), often understood as a proxy measure of income/SES, and race or ethnicity (n = 16). Many studies also specified additional social disadvantage variables of interest, which most commonly included income/SES (n = 32), race/ethnicity (n = 32), level of education (n = 17), sex/gender (n = 16), rural/remote location (n = 16), and age (n = 15) (see Table 3). However, despite many studies identifying more than one social disadvantage variable of interest, none explicitly identified using an intersectional approach in their methodology or conceptual understanding of social disadvantage.

Interventions, services and models of care

Interventions, services and models of care ('interventions') were designed to address a range of barriers. The most described barriers addressed were access to health insurance, financial barriers, and transportation barriers (Table 4). Interventions spanned a range of implementation settings, including single- and multi-center settings, as well as interventions implemented on a much broader scale, most notably, the policy-based expansion of health insurance ('Medicaid expansion') in the United States (Table 4). The most common interventions evaluated were policy-focused, particularly Medicaid expansion in the U.S., followed by patient navigation interventions. Education-focused interventions were relatively few, and their impact appeared limited, while institutional and system-level interventions showed more consistent improvements in access. Interventions were categorized and are described according to their primary focus: micro level interventions (primarily focused on individual patients and/or service providers; n = 4), micro-meso level interventions (navigation interventions; n = 14), meso level interventions (primarily targeting

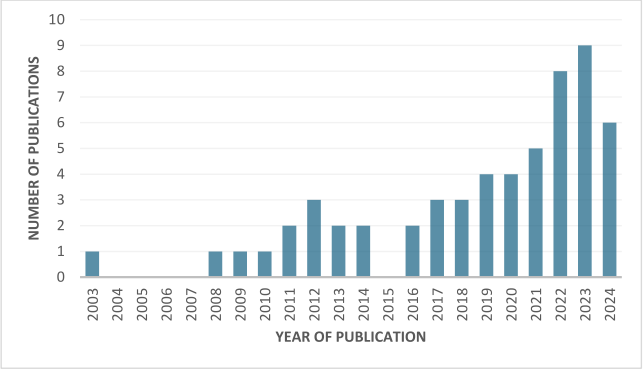


Fig. 2. Year ofpublication.

Table 3
Study characteristics.

Study Characteristic	Number of Studies (n) (proportion of studies as %, if applicable)
Study Design	
Observational	33 (58%)
Quasi-experimental	13 (23%)
Randomized controlled trial	4 (7%)
Quality improvement	5 (9%)
Pragmatic trial	2 (3%)
Cancer Focus	
Single type of cancer	31 (54%)
Multiple types of cancer	20 (35%)
Not specified	4 (7%)
Non-specific	2 (4%)
*Type of Cancer**	
Gastrointestinal (GI)	28
Breast	27
Genitourinary (GU)	19
Lung	15
Gynecological	10
Liver	10
Head & Neck	7
Melanoma	6
Sarcoma	2
Brain & CNS	4
Hematologic/Blood	4
Other	2
*Type of Cancer Treatment**	
Surgery	37
Systemic therapy	27
Chemotherapy	21
Hormone therapy	8
Immunotherapy	5
Targeted therapy	1
Not specified	4
Radiotherapy	25
Therapeutic clinical trial	10
Not specified	4
Bone marrow transplant	1
*Primary Type of Social Disadvantage of Interest (as reported by authors)***	
Insurance coverage (as a proxy for SES)	18
Race or ethnicity	16
Rural/remote geographical location	2
Vulnerable or underserved	2
Uninsured or underinsured	1
Additional Types of Social Disadvantage of Interest (as reported by authors)	
Race or ethnicity	32
Income/socioeconomic status	32
Education level	17
Sex/gender	16
Rural/remote location	16
Age	15
Access to insurance	10
Facility/residence type	10
Employment status	4
Marital status	3
Cultural/linguistic	2
Cultural/linguistic	2
Mental health diagnosis	1
Housing	1
Religion	1

General Note: Totals may exceed the number of included studies because some studies reported multiple categories.
* Totals may exceed 57 as some studies included multiple cancer types or treatments. ** Totals may be less than 57 because only 40 studies specified a singular primary variable of interest; primary and additional variables are reported separately where applicable.

Table 4
Characteristics of interventions, services and models of care.

Characteristic of Intervention	Number of studies (n)	Proportion of studies (%, if applicable)
Type of Intervention* †		
Education-focused	4	7%
Patient navigation focused	14	25%
Institutional and system-focused	7	12%
Policy-based insurance interventions	32	56%
Type of Barriers Addressed (as reported by authors)*		
Access to insurance	35	27%
Age	2	2%
Cultural/linguistic	6	5%
Discrimination	8	6%
Geographic	4	3%
Housing	1	1%
Financial	27	21%
Health/mental health	2	2%
Health literacy	10	8%
Racial/ethnic	10	8%
Social support	5	4%
System navigation	9	7%
Transportation	11	8%
Setting of Intervention and/or Evaluation*		
National	10	18%
Multiple states	17	29%
Single state	5	9%
Region within a state	2	3%
Multi-city	1	2%
Single center/site	14	25%
Multiple centers/sites	6	11%
Not specific	2	3%

*Total Studies n = 57
† In some studies, the intervention was implemented on a larger scale than was evaluated. E.g., Medicaid expansion was implemented across many states, but some studies reported only a single- or multistate-based analysis.
**Total barriers addressed n = 130 (130 barriers were identified across 57 studies, as some studies addressed multiple barriers).

institutional or health system changes; n = 7), and macro level interventions (focused on broader policy structures; n = 32).

Micro level: education-based interventions

Four studies described interventions that primarily targeted individuals through education-based initiatives (Table 5); three of these were focused on patient education (Fracasso et al., 2013; Robinson et al., 2017; Skinner et al., 2019) and one on healthcare provider education (Wells et al., 2017). Educational strategies included cultural competency training, printed and video patient educational materials, and an individualized coaching intervention primarily focused on patient education. Evaluations of these interventions suggest that educational interventions had little impact on improving access to cancer treatment, with only one of four studies showing a positive impact on enrollment in therapeutic clinical trials (Robinson et al., 2017).

Micro-meso level: patient navigation interventions

Fourteen studies described interventions that focused on patient navigation services to improve access to cancer treatment (Table 5), straddling micro and meso-level interventions. This included a mix of patient navigation services delivered by lay navigators (i.e., non-professional, non-clinical personnel but had training specific to cancer patient navigation) (Battaglia et al., 2022; Fouad et al., 2016; Ramirez et al., 2014; Vilchis et al., 2019), professional navigators (i.e., trained healthcare providers specializing in cancer patient navigation, often oncology nurses) (Goodwin et al., 2003; Holmes et al., 2012; Hunnibell et al., 2012; Proctor et al., 2011), or a combination of both (Dudley et al., 2012a); the remaining studies did not specify whether navigators were lay or professional. Patient navigation services varied in terms of timing: more than half of the studies reported providing patient navigation through to the completion of cancer treatment (Battaglia et al., 2022, 2022; Chen et al., 2010; Dudley et al., 2012a; Fouad et al., 2016; Guadagnolo et al., 2011; Holmes et al., 2012; Hunnibell et al., 2012); fewer provided navigation services between the time of cancer diagnosis to treatment initiation (Proctor et al., 2011; Schwaderer et al., 2008; Vilchis et al., 2019). Patient navigation also varied in terms of service delivery with some patient navigators embedded within tertiary care centers (e.g., public hospitals) or comprehensive cancer centers, and others embedded in community cancer centers or communities themselves (see Table 5).

Patient navigation services commonly involved elements of (a) healthcare system-based navigation such as care coordination and liaising between care providers; and (b) patient-based navigation, including proactive identification of barriers to care and individualized care plans to mitigate such barriers; and individualized patient education and emotional support. Notably, all patient navigation interventions addressed multiple barriers, often at both the system and individual levels. Several patient navigation services were specifically designed and delivered to be culturally tailored, and some of these included a component of community outreach and/or engagement, often by lay community health workers. One study reported on a nurse-facilitated case management intervention that included many of the components of patient navigation, but without any system-based components (e.g., monitoring timelines for receipt of treatment) (Goodwin et al., 2003).

Evaluations of patient navigation interventions were mixed, with 3 studies showing no impact on access to treatment and two studies showing some impact on access to treatment (although Battaglia et al. 2022 compared enhanced sociolegal navigation services with standard navigation services; findings indicated that enhanced sociolegal navigation services did not have additional benefit in improving access to cancer treatment). The remainder of studies (n = 9) demonstrated a positive impact on access to treatment, including improved receipt of treatment, improved adherence to quality treatment indicators, and reductions in treatment delays, interruptions, and barriers to care. Notably, several studies reported that patient navigation services demonstrated a reduction in access to treatment disparities for socially disadvantaged groups, and although Vilchis et al. (2019) reported no improvement in access to treatment, they did find a positive impact on patient and caregiver satisfaction with care.

Meso-level: Institutional and system-focused interventions

Seven studies described interventions aimed at strengthening the health system or healthcare institutions within the health system, to improve access to cancer treatment (Table 5). These interventions varied in terms of their components, with five interventions describing multiple components meant to work synergistically (Anwuri et al., 2013; Charlot et al., 2022; Cykert et al., 2020; Nipp et al., 2019; Tang et al., 2021). For example, Nipp et al. (2019) reported on a multi-component intervention (the Cancer Care Equity Program), which included community outreach and education, patient navigation, and a financial assistance program to improve access to clinical trials. Five studies described strategies to create, change, or improve institutional processes and structures; this included creating information feedback loops to healthcare providers, reminiscent of a learning health systems approach (Anwuri et al., 2013; Charlot et al., 2022; Cykert et al., 2020; Tang et al., 2021); creating a weekly tumor board to bring interdisciplinary cancer providers together with the goal of streamlining care and reducing disparities (Tang et al., 2021); and implementing referral scheduling structures to improve timeliness of care (Chen et al., 2023). Three studies reported on interventions that paired patient navigation with other intervention components; of these, two studies described a model of patient

Table 5
Selected data charting.

Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
Education-Based Interventions					
Fracasso et al. (2013)	Self-reported race and ethnicity/ Gender, socioeconomic (education, employment and family status)	Advanced breast, colorectal, lung, or prostate carcinoma	Clinical trials (not otherwise specified)	Coach Intervention (CI): a lay coach to provide flexible, individualized, nondirective basic education and support for patients, creating a trusting context that promotes clinical trial enrollment.	CI had no impact on adherence or clinical trial enrollment compared with usual care, and contact frequency did not affect outcomes. Higher baseline QOL (FACT-G) predicted greater trial enrollment, while concerns about research risks for minorities and low-income groups predicted lower enrollment. These findings suggest that improving patient perceptions and QOL may be more effective than coaching alone in boosting trial participation.
Robinson et al. (2017)	Self-reported race / Sociodemographic variables, income, education, and religion.	Breast cancer	Clinical trials (not otherwise specified)	A 15-minute culturally tailored educational video designed to address six key barriers to clinical trial participation, including mistrust, fear of unfair treatment, and lack of knowledge. Participants completed questionnaires before, immediately after, and 7–21 days after viewing the video to measure changes in attitudes and intentions. Study coordinators provided follow-up support for six months to encourage and track enrollment in available therapeutic clinical trials.	The video intervention significantly improved Black breast cancer patients' willingness to participate in clinical trials, with trial enrollment increasing from a baseline rate of 6–13.5% within six months. Participants showed more positive attitudes on 81% of measured barriers, including greater.
Skinner et al. (2019)	Race and ethnicity/ Education, age, sex	Urology, hematology and breast cancers	Clinical trials (not otherwise specified)	A 10-minute educational video intervention delivered in the clinic to oncology patients, designed to explain clinical trials, potential benefits and risks, and include patient advocate stories and oncologist interviews. Participants were randomized to either watch the video in the office (intervention group) or receive the same video and booklet to review at home (control group). All participants completed a pre-survey, a follow-up phone survey one week later, and their clinical trial enrollment was tracked for one year.	The study found no statistically significant difference in clinical trial enrollment between the intervention and control groups. The video also did not significantly improve clinical trial knowledge or reduce perceived barriers to participation. Minority participants consistently reported more negative expectations and higher barriers to participation than White participants, highlighting the need for more tailored, culturally relevant strategies to address mistrust and information needs in underrepresented groups.
Wells et al. (2017)	Race and ethnicity (Latino and African American populations)	Not specific	Clinical trials (not otherwise specified)	A 4-hour Cultural Competency and Recruitment Training Program (CCRTP) for clinical research associates and physicians, focusing on cultural barriers and strategies to improve minority enrollment in radiation therapy clinical trials.	The training enhanced participants' cultural awareness and attitudes, resulting in a 1.5% increase in minority enrollment at participating sites—approximately 300 additional patients in one year. Though not statistically significant, the program showed promise for reducing disparities in clinical trial access.
Patient Navigation Interventions					
Battaglia et al. (2022)	Sociolegal barriers/ Housing and employment barriers	Breast cancer and non-small cell lung cancer	Surgery; Radiotherapy; Systemic therapy	Patient-Navigation, comparing enhanced sociolegal navigation with standard patient navigation. It included 3 steps:	High rates of timely treatment (96% for breast, 73% for lung) were observed in both groups; however, enhanced navigation

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Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
			(Chemotherapy and Hormone therapy)	1. Comprehensive sociolegal needs assessment, 2. Longitudinal interdisciplinary care planning by the navigator in consultation with a lawyer, 3. Referral to direct legal support Patient navigation (Project SUPPORT) at Boston Medical Center (2013–2017) provided by nonclinical or lay navigators to assess barriers, plan care, and track patients; the enhanced arm included legal partnership to address sociolegal needs. Services began within 30 days of diagnosis and continued through treatment.	showed no added effect on access to treatment. Most patients (75%) faced sociolegal barriers, but only a few required direct legal aid. While legal-enhanced navigation did not improve outcomes beyond standard navigation, the program showed that systematic documentation and support for sociolegal needs are feasible in safety-net settings and contribute to addressing health equity in cancer care.
Chen et al. (2010)	Socioeconomic status, financial, underinsured, race and ethnicity/ cultural and linguistic.	Breast cancer	Surgery; Radiotherapy; Systemic therapy (Chemotherapy; Hormone therapy)	A breast cancer patient navigator program was implemented in August 2006 at a Los Angeles County public hospital. Bilingual, bicultural navigators (English/Spanish) trained in case management, medical interpretation, and cultural competency provided outreach, health education, psychosocial support, and care coordination. They worked one-on-one with patients from diagnosis through completion of acute treatment, assisting with transportation, financial resources, and continuity of care.	Among 100 newly diagnosed breast cancer patients (Stages I–III), adherence to 9 ASCO quality-of-care indicators improved from 69% before navigation to 86% after. The program reduced gaps in quality of care for uninsured, predominantly Hispanic and Spanish-speaking patients.
Dockery et al. (2018)	Race -Being an American Indian as determined based on electronic health information	Cervical cancer	Surgery; Radiotherapy; Systemic therapy (Chemotherapy)	The American Indian (AI) -specific patient navigation program established in 2005 at the Stephenson Cancer Center, University of Oklahoma (Oklahoma City, USA). Navigators acted as liaisons between tribal clinics, Indian Health Service (IHS) facilities, and the cancer center to assist with referrals, authorizations, and access to financial and social resources. Navigation began after referral and supported patients through diagnosis and treatment.	Navigation did not significantly reduce time to treatment initiation (30.5 vs. 27.5 days) or improve treatment completion (92% vs. 100%). Systemic IHS referral and funding barriers limited impact. Navigation may still offer less tangible benefits such as support and improved coordination.
Dudley et al., (2012a)	Race-ethnicity, language, country of origin/ Income (insurance status, employment, marital status)	Breast cancer	Not specified	Combined patient navigator/ promotora model implemented at the University Health System and the University of Texas Health Science Center at San Antonio (UTHSCSA), Texas, between 2006–2010. The program paired professional navigators (nurses, social workers, public health staff) with promotoras (lay community health workers) to address both medical and cultural barriers for predominantly low-income Hispanic women with abnormal breast screenings or untreated breast cancer. Navigation began after abnormal screening or diagnosis and continued until diagnosis resolution or completion of initial therapy.	Navigation significantly reduced the time from diagnosis to treatment initiation (57 vs. 74 days, $p = 0.04$), especially among Hispanic women (56 vs. 81 days, $p = 0.02$). More navigated women began treatment within 60 days of diagnosis (80% vs. 56%). The program narrowed racial/ ethnic disparities in timely breast cancer care and improved equity in access to diagnosis and treatment.

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Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
Fouad et al. (2016)	Race - identify African American/ education, income, health insurance status, and employment status	Brain and nervous system, female breast, lung, lymphoid leukemia, myeloid and monocytic leukemia, ovary, skin-melanoma, non-Hodgkin lymphoma prostate. kidney	Clinical trials (not otherwise specified)	The Increasing Minority Participation in Clinical Trials (IMPaCT) project was implemented at the University of Alabama at Birmingham Comprehensive Cancer Center (USA) between 2007–2014. Lay community members were trained as patient navigators to educate African American cancer patients about clinical trials and assist with recruitment and retention. Services included education, transportation and lodging support, reminders, linkage to community resources, and culturally tailored emotional support. Navigation began at the point of referral or trial offer and continued through trial participation.	The program increased African American participation in therapeutic cancer clinical trials from 9% to 16% and doubled retention rates (74.5% with navigation vs. 37.5% without). It effectively addressed racial disparities in trial participation by integrating navigators into clinical teams and mitigating both system- and patient-level barriers.
Goodwin (2003)	Older age women/ Ethnicity (minority representation), income, medical insurance, education), and mental health.	Breast cancer	Surgery; Radiotherapy; Systemic therapy (Chemotherapy; Hormone therapy)	Nurse case managers provided personalized support to women aged 65 + newly diagnosed with breast cancer, starting within 24 h of identification. Services were delivered over 12 months across 15 hospitals in southeast Texas, through home visits, phone calls, and accompaniment to medical appointments. The intervention focused on assessment, planning, implementation, and evaluation to improve care coordination and patient understanding.	Increased use of guideline-based treatments (e.g., radiation, breast-conserving surgery), improved satisfaction and arm function. Significantly, the intervention narrowed disparities in treatment among high-risk groups—such as women who were older, unmarried, living alone, or of minority ethnicity—by improving their likelihood of receiving appropriate care. This suggests that nurse case management can be a powerful tool for promoting health equity in cancer treatment for vulnerable populations.
Guadagnolo et al. (2011)	Race (in particular being an American Indian) was self-identified by patients/	Any malignancy	Radiotherapy; Clinical trials (not otherwise specified)	Culturally tailored patient navigation was provided to American Indian cancer patients at Rapid City Regional Hospital in South Dakota, starting at diagnosis and continuing throughout treatment. Navigators offered in-person and phone support to coordinate appointments, assist with insurance, medications, transportation, lodging, and provide psychosocial support. Services began at first visit and extended across tribal communities in the Northern Plains region.	Navigation significantly reduced radiation treatment interruptions (mean 1.7 vs 4.9 days) and led to higher clinical trial enrollment (22%) among a historically underserved population. The program improved continuity of care and addressed logistical and cultural barriers, promoting equity in cancer treatment delivery.
Holmes et al. (2012)	Race	Breast cancer	Clinical trials (not otherwise specified)	At the University of Southern California Norris Comprehensive Cancer Center (Los Angeles, CA), an oncology nurse navigator program was implemented from 2007 to 2008. The navigator worked directly in community physician offices in underserved areas, service planning areas (SPA) 6, engaging newly diagnosed Black breast cancer patients. Services included education, emotional support, trial screening, consent assistance, and coordination of logistics (appointments,	The program significantly increased clinical trial participation among Black patients—86% of eligible patients enrolled, raising overall Black enrollment from 3% to 7%. It reduced geographic and systemic barriers, strengthened trust between patients and providers, and demonstrated a cost-effective model for improving minority representation in cancer research.

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Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
				transportation, medical records), starting at diagnosis and continuing through trial enrollment and follow-up.	
Hunnibell et al. (2012)	socioeconomic status affecting timeliness include race and education.	Lung cancer	Not specified	At the Connecticut Veterans Affairs Healthcare System (CT-VAHCS), an advanced practice nurse was hired in 2007 as a cancer care coordinator to serve as a nurse navigator for veterans with suspected lung cancer. The coordinator engaged patients from the point of abnormal imaging through diagnosis and initiation of treatment, streamlining referrals, staging studies, and multidisciplinary coordination across radiology, pulmonary, and oncology departments.	The intervention reduced time from suspicion to treatment from 136 days (2003) to 52 days (2010) and increased early-stage (I/II) lung cancer diagnoses from 33% to 53%. It improved care timeliness for a high-risk, underserved veteran population, addressing systemic delays and enhancing equity in cancer care delivery.
Ko et al., (2014)	low-income, uninsured, or publicly insured patients/ racial and ethnic minority groups	Breast cancer	Surgery; Radiotherapy; Systemic therapy (Chemotherapy; Hormone therapy)	As part of the National Patient Navigation Research Program (PNRP), eight U.S. sites implemented patient navigation between 2006–2011 to support underserved women with newly diagnosed breast cancer (specific model not well described). Navigators—nonclinical staff trained in cancer care—guided patients from diagnosis through treatment, helping overcome barriers such as insurance, transportation, language, and system complexity. Services were delivered in community health centers and safety-net hospitals across cities including Boston, Chicago, Denver, San Antonio, and Washington, DC.	Navigated patients were significantly more likely to receive antiestrogen therapy (OR 1.73; P = .004), a key standard of care for hormone receptor–positive breast cancer. While no significant improvement was seen for radiation or chemotherapy initiation, the program demonstrated that navigation can improve adherence to recommended treatment among vulnerable populations, helping reduce disparities in breast cancer care.
Proctor et al. (2011)	Under- or uninsured status, age 80 years, membership in a racial/ ethnic minority, receipt of medical assistance, and residence in economically deprived areas	Glioblastoma, non-small-cell lung cancer; breast, head and neck cancer	Radiotherapy	At the University of Pittsburgh Medical Center (UPMC)/ Jameson Cancer Center in New Castle, Pennsylvania, a patient navigator was integrated into a community radiation oncology disparities program during a 9-month study period. The navigator worked alongside clinic staff to identify and support patients—especially those at risk for disparities (e.g., uninsured, elderly, minority)—from initial consult through trial enrollment. Services included education, barrier reduction, and coordination of consent and trial logistics.	Use of a screening tool and navigator services led to high clinical trial enrollment rates among eligible patients, with similar accrual rates for both disparity and non-disparity groups. The navigator helped ensure no eligible patients were missed, demonstrating that navigation can effectively support trial participation and equity in access at community cancer centers.
Ramirez et al. (2014)	Race/ Transportation and/or child care, imprecise communication with health care providers, health insurance issues, and fear of cancer and/or treatment of it.	Breast cancer	Surgery; Radiotherapy; Systemic therapy (chemotherapy, hormone therapy)	From 2008–2011, the Redes En Acción network implemented a culturally tailored patient navigation program for Latinas with breast cancer across six U. S. cities (San Antonio, Houston, San Diego, San Francisco, Miami, and New York). Trained bilingual Latina navigators provided personalized support from diagnosis to treatment initiation, including appointment accompaniment, transportation coordination,	Navigated Latinas were significantly more likely to begin treatment within 30 days (69% vs. 46%) and 60 days (98% vs. 73%) of diagnosis. The intervention reduced average time-to-treatment by over 26 days and helped overcome cultural, linguistic, and logistical barriers, improving timely access to care for an underserved population.

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Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
				emotional support, translation, insurance assistance, and family engagement.	
Schwaderer et al. (2008)	Disparities, underinsurance/ Age and race	Not specified	Radiotherapy	At Jameson Cancer Center in New Castle, Pennsylvania, a patient navigation program was implemented in 2005 as part of a National Cancer Institute disparities initiative. Navigators identified underserved patients—based on insurance status, race, age, or housing—and provided support from diagnosis through treatment planning. Services included coordination with medical staff, barrier identification, and limited follow-up during radiation therapy unless additional needs arose.	Navigated patients began treatment slightly faster than un-navigated groups (average 21 vs. 23–25 days) but had lower treatment completion rates and more missed appointments. While navigation showed potential to reduce delays, its limited involvement during treatment may have affected outcomes. The study highlights the need for continuous navigator support to improve equity in cancer care delivery.
Vilchis et al. (2019)	Rurality/ language, gender, age, education, ethnicity and income	Not specified	Not specified	The Una Mano Amiga pilot program was implemented from 2008 to 2012 in three rural counties of southwest New Mexico (Grant, Luna, and Hidalgo). Two bilingual promotoras were trained as patient navigators to assist newly diagnosed cancer patients and their families. Services included coordinating care, arranging referrals, addressing barriers (e.g., financial, social, insurance, fear), and providing education and emotional support through phone and in-person encounters.	The intervention did not reduce time from diagnosis to treatment compared to historical data (i.e., no impact on timeliness of treatment), but patients and families reported high satisfaction. Navigators successfully addressed key barriers and reduced caregiver stress, demonstrating the model's value in supporting underserved rural populations and improving patient experience.
Institutional and System-Focused Interventions					
Anwuri et al. (2013)	Race (African Americans and others, including American Indians/Alaska Natives, Asians, and Native Hawaiians or other Pacific Islanders)/ Disease-specific accrual targets and incidence rate by race and gender	Leukemia/lymphoma, breast, gynecologic, gastrointestinal (lower and upper), pediatric cancers, thoracic, genitourinary, head and neck, endocrine, melanoma, musculoskeletal, neuro-oncology.	Clinical trials (not otherwise specified)	A system-level intervention to increase minority participation in therapeutic clinical trials. The multi-component navigation model (program) included institutional support, policy changes, and infrastructure to monitor and guide accrual. The Program for the Elimination of Cancer Disparities (PECaD) reviewed trial protocols, set race-specific accrual benchmarks, and provided feedback and support to investigators throughout the trial lifecycle.	Minority enrollment in therapeutic trials increased from 12% in 2005–14% in 2010, with a 60% rise in minority participation across all trials. Although proportional representation remained below the minority cancer incidence rate (17.5%), the structured approach improved accountability and helped close gaps in trial access for underserved populations.
Charlot et al. (2022)	Race - as recorded in a patient's EHR/ Other patient demographics (income, age, sex, residence)	Lung cancer	Surgery	A multi-component system-based intervention consisting of: (a) a real-time warning system derived from automated uploads from electronic health records delivering alerts when patients missed a scheduled appointment or did not reach an expected milestone in care (e.g., no surgery or radiotherapy received within 90 days); (b) a nurse navigator who accessed the warning system daily, re-engaged with patients directly to identify and address barriers to care (e.g., logistical, financial, mistrust, cultural, miscommunication), and then	The intervention significantly improved timely surgery rates for both Black and White patients with early-stage lung cancer. Black patients in the intervention group were 43% more likely to receive surgery within 8 weeks compared to historical controls, effectively eliminating the racial gap in timely treatment. The study demonstrated that antiracism system-based navigation can enhance equity and overall quality in cancer care delivery.

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Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
				engaged with physicians to address treatment milestone issues, and noted resolutions in the warning system; and (c) a physician champion (practicing oncologist) at each center who continually reminded staff of the study, and provided quarterly feedback reports for the cancer center on treatment completion stratified by race.	
Chen et al. (2023)	Vulnerable with focus on race (Blacks, Latinos, and Asian patients)/ Health disparities, insurance status.	Breast, genitourinary, gastrointestinal, head-and-neck, and lung; central nervous system, gynecologic, lymphoma, soft tissue and skin; and benign/others.	Radiotherapy	A same-day access initiative (appointment scheduling intervention) was implemented in the radiation oncology department. Intake staff offered newly referred patients, once insurance authorization was confirmed, the option to be seen by a radiation oncologist that same day. The system included real-time scheduling, direct communication with providers, and personalized coordination of records and logistics to streamline access.	The initiative significantly improved access for Black, Latino, and Asian patients, who were more likely to use same-day appointments than White patients. It reduced no-show rates (from 20% to 7% for Black patients and 14–5% for Latinos) and shortened time from referral to consultation, especially for underserved groups. These changes narrowed racial disparities in timely access to radiation oncology care.
Cykert et al., (2019)	Race/ Income (health insurance) and residency	Lung cancer	Surgery; Radiotherapy	A multi-component system-based intervention consisting of: (a) a real-time warning system delivering alerts when patients missed a scheduled appointment or did not reach an expected milestone in care; (b) a nurse navigator who accessed the warning system daily, re-engaged with patients directly to identify and address barriers to care, engaged with physicians to address treatment milestone issues, and noted resolutions in the warning system; and (c) a physician champion at each center who engaged staff and provided quarterly feedback reports for the cancer center on treatment completion stratified by race.	The intervention eliminated racial disparities in receipt of curative treatment. Black patients' treatment rates increased from 69% (retrospective) to 96.5%, matching White patients. Logistic regression confirmed parity in surgery and radiation use. The system improved care for all patients and demonstrated that race-specific transparency and real-time navigation can close longstanding gaps in cancer treatment.
(Nipp et al., 2016)	Financial	Not specified	Clinical trials (not otherwise specified)	The Cancer Care Equity Program (CCEP) was implemented in 2013 to reduce financial barriers to clinical trial participation. Multi-component intervention (program) community outreach included patient navigation and financial assistance for travel and lodging, provided in partnership with the Lazarex Cancer Foundation. Patients were referred during trial screening or enrollment and reimbursed based on income level.	Following CCEP implementation, clinical trial enrollment increased significantly. Participants were more likely to be younger, female, lower-income, live farther from the hospital, and report financial concerns. The program helped enroll patients facing greater financial burden, highlighting the role of navigation and financial support in improving equity in trial access.
Scrace and Margolis, 2009	Rurality	Skin cancer (melanoma and non-melanoma skin cancer)	Surgery	The Royal Flying Doctor Service piloted a fly-in/fly-out skin cancer clinic targeting remote communities. A second physician with expertise in skin cancer care joined regular primary care visits, focusing exclusively on skin checks, diagnosis, and treatment. Services were delivered onsite	The clinic reached 29% of the adult non-Indigenous population, with improved melanoma detection—particularly among older men—rising from 0.2 to 2 per 1000 people/year. Diagnostic accuracy matched metropolitan standards, demonstrating that targeted outreach can enhance cancer

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Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
				using existing infrastructure and resources to minimize costs	care equity in geographically isolated populations.
Tang et al. (2021)	Underserved patients at safety net hospital/ Race and type of insurance	Breast cancer	Surgery; Systemic therapy (Chemotherapy)	A multidisciplinary (multi-component) breast program to improve breast cancer care for underserved patients. The intervention included co-located breast and oncology clinics, a weekly tumor board, and standardized diagnostic and treatment protocols. While not a traditional navigation model, the program aimed to streamline care coordination and reduce delays.	There was no significant improvement in the time to imaging, biopsy, or treatment initiation, or in the timeliness of treatment. However, care delivery was equitable across racial groups within the same socioeconomic status. The study highlighted systemic delays and the need for targeted interventions to improve timeliness and outcomes in low-resource settings.
Policy-Focused Interventions					
Adamson et al., (2021)	Insurance coverage (as a proxy measure of SES)/ Race	Advanced or metastatic solid tumors, including non-small cell lung cancer, colorectal cancer, breast cancer, gastric/ esophageal cancer, melanoma, renal cell carcinoma, prostate cancer	Systemic therapy (not otherwise specified)	The Affordable Care Act (ACA), a policy-based healthcare reform, expanded Medicaid eligibility (public health insurance) to those with incomes of up to 138% of the federal poverty line. The goal of the ACA was to improve access to healthcare nationwide for low-income Americans. Implementation of the act was at the discretion of each state – states could choose to implement this ('Medicaid Expansion States') or not ('non-expansion states'). The ACA was signed into law in 2010; the major provisions of the ACA were implemented in 2014.	ACA Medicaid expansion reduced racial disparities in timely cancer treatment, nearly eliminating the gap between Black and White patients in receiving therapy within 30 days of diagnosis.
Ajkay et al., (2018)	Insurance coverage (as a proxy measure of SES)/ Income, race, education	Breast cancer	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA), as described above.	Medicaid expansion in Kentucky led to increased insurance coverage, earlier-stage breast cancer diagnoses, and greater use of breast-conserving surgery. It improved access and treatment appropriateness, though gains in treatment timeliness were modest.
Albright et al. (2021)	Women without insurance/Income, race, ethnicity, facility/ residence type, education	Gynecologic cancers (cancers of the uterus, ovary and fallopian tube, cervix, vagina, and vulva)	Surgery, Radiotherapy, Systemic therapy (not otherwise specified)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion led to reduced uninsurance rates, increased early-stage diagnoses, and improved access to timely treatment (i.e., timeliness of treatment) for non-elderly women with gynecologic cancers, especially benefiting low-income, Hispanic, and cervical cancer patients.
Al-Refaie et al., (2017)	Insurance coverage (as a proxy measure of SES)/ racial and ethnic minorities	Lung, esophageal, stomach, liver, pancreas, colon, rectum, kidneys, or urinary bladder	Surgery	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion significantly reduced uninsured cancer surgeries and increased Medicaid-covered procedures, shifting the financial burden from patients to Medicaid. However, it did not increase overall surgical volume or improve racial equity in access.
Crocker et al., (2019)	Racial and ethnic minorities/income, insurance status	Colorectal, esophagogastic, pancreatic	Surgery	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion led to a significant increase in cancer surgery utilization among low-income patients but did not improve racial disparities. Coverage gains translated into better access for economically

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Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
					disadvantaged groups, highlighting persistent barriers for racial minorities.
Desai et al., (2020)	Income, education, insurance coverage/ racial and ethnic minorities	Renal cell carcinoma	Surgery, Systemic therapy (chemotherapy, immunotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion did not significantly improve insurance coverage or timely treatment for patients with metastatic renal cell carcinoma, suggesting limited impact on access or equity for this cancer type.
Eguia et al., (2018)	Insurance coverage (as a proxy measure of SES)/ Access to insurance, income, race	Pancreatic, esophageal, lung	Surgery	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion led to increased cancer admissions and surgical resections, especially for lung, breast, and colorectal cancers—and narrowed disparities in access among Medicaid patients and racial minorities.
Elshami et al., (2022)	Insurance coverage (as a proxy measure of SES)/ Education, income, race	Hepatocellular carcinoma (HCC)	Surgery	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion was associated with earlier diagnosis, increased rates of resection and transplantation (i.e. increased access to surgical treatment), and improved survival for patients with HCC, especially those with early-stage disease.
Fonseca et al., (2022)	Insurance coverage (as a proxy measure of SES), state of residence/ Income, education	Pancreatic adenocarcinoma (PDAC)	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion was associated with increased access to curative surgery, multimodal therapy, and treatment at high-volume centers for pancreatic cancer patients, along with reduced short-term mortality and improved overall survival.
Gaba (2023)	Self-reported race/Access to insurance, income	Breast cancer	Surgery, Systemic therapy (chemotherapy, immunotherapy, hormone therapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion reduced uninsured rates and slowed declines in timely treatment overall but did not improve racial disparities in stage at diagnosis, treatment initiation, or mortality between American Indians and Whites.
Hoehn et al., (2021)	Income, age, insurance coverage (as a proxy measure of SES)	Colon cancer	Surgery	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion led to earlier-stage diagnoses, faster treatment for early-stage patients, and increased use of minimally invasive surgery (enhanced access to surgical care for colon cancer).
Irace et al., (2023)	Income, insurance coverage (as a proxy measure of SES)/Race or ethnicity, education, and residence	Rhinologic cancer	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion reduced uninsured rates and was associated with fewer advanced stage diagnoses but did not improve survival or treatment rates (no impact on receipt of treatment). Uninsured patients had significantly worse outcomes, highlighting persistent disparities in access and prognosis.
Janopaul-Naylor et al. (2023)	Income, insurance coverage (as a proxy measure of SES)/Race, ethnicity	Prostate cancer	Surgery, Radiotherapy, Systemic therapy (immunotherapy, hormone therapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Despite expanded coverage, Medicaid patients faced similar treatment delays as the uninsured, and racial disparities in timely care persisted. Treatment delays increased overall after ACA implementation, suggesting insurance gains did not translate into improved access.

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Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
Jiang et al., (2023)	Race, ethnicity/Income, access to insurance	Bladder cancer	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion significantly reduced racial disparities in timely multidisciplinary treatment for muscle-invasive bladder cancer, improving access for Black patients in expansion states compared to non-expansion states.
Kaelberer et al., (2023)	Income, insurance coverage (as a proxy measure of SES)/ Geographical location, race	Hepatopancreatobiliary (HPB) and gastrointestinal (GI) cancers, including liver cancer, pancreatic cancer, colorectal liver metastasis and gastric cancer	Surgery, Systemic therapy (not otherwise specified)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion did not significantly increase surgical resection rates for liver, pancreatic, gastric, or colorectal liver metastases, though more patients in expansion states received care at academic centers.
Laughlin et al. (2023)	Insurance coverage (as a proxy measure of SES)/ Age, race, geographical location, income	Breast cancer	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion in Southern states led to increased insurance coverage and earlier-stage diagnosis among low-income women. While more patients in expansion states received treatment (surgery, chemotherapy, radiation), this increase was not directly attributable to Medicaid expansion itself. The authors found no significant effect of the policy on treatment receipt, suggesting other state-level factors may be driving improved access.
Lee et al., (2021)	Insurance coverage (as a proxy measure of SES)/ Race, income, facility type, geographical location,	Cervical cancer	Surgery, Radiotherapy, Systemic therapy (Chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion increased insurance coverage for cervical cancer patients but did not significantly affect stage at diagnosis, timely treatment, or survival. The benefits of expanded coverage may require longer follow-up to assess downstream effects.
Lima et al. (2023b)	Insurance coverage (as a proxy measure of SES)/ Age, gender, race/ ethnicity, income, facility type	Intrahepatic Cholangiocarcinoma (ICC)	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion improved early-stage diagnosis, surgical treatment rates, and survival for ICC patients overall. However, racial and socioeconomic disparities persisted, with non-Hispanic Black and lower-income patients less likely to receive surgery or multimodal care.
Lima et al., (2023a)	Insurance coverage (as a proxy measure of SES)/ Race, gender, income, education, facility type, geographical location	Hepatocellular carcinoma (HCC)	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion increased surgical treatment rates and improved survival among uninsured/Medicaid patients with early-stage HCC. However, overall surgery rates declined post-expansion, and benefits were not observed among privately insured or other groups, highlighting uneven gains in access.
(Lin et al., 2022)	Insurance coverage (as a proxy measure of SES)/ Age, sex, race, income, facility type, geographical location	Colorectal cancer	Surgery, Systemic therapy (not otherwise specified)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion led to greater use of neoadjuvant chemoradiotherapy (NCRT) and increased the likelihood of receiving surgical treatment.
Liu et al., (2020)	Insurance coverage (as a proxy measure of SES)/ Age, race/ethnicity, marital status	Prostate cancer	Surgery, Radiotherapy	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion increased insurance coverage among men with high-risk prostate cancer but did not improve treatment rates. Patients with Medicaid

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Table 5 (*continued*)

Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
					were no more likely to receive definitive therapy than uninsured patients, highlighting persistent disparities in access and care quality.
Mesquita-Neto et al., (2020))	Insurance coverage (as a proxy measure of SES)/ Age, sex, race, income	Breast, colorectal, ovarian, lung, uterine, pancreatic, prostate, and liver cancers	Surgery	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion reduced the uninsured rate, led to earlier cancer diagnoses, and significantly improved access to surgery for patients from lower socioeconomic backgrounds. However, racial disparities in surgical access persisted, with African-American and Hispanic patients less likely to receive recommended treatment.
Mobley et al., (2023))	Insurance coverage (as a proxy measure of SES)/ Age, sex, level of education, geographic location, income, race	Gastrointestinal (GI) cancers (anus, colon, esophagus, rectosigmoid junction, liver, pancreas, rectum, small intestine and stomach)	Surgery, Radiotherapy, Systemic therapy (chemotherapy, immunotherapy, hormonal therapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion was associated with improved timely treatment initiation for gastrointestinal cancer patients, especially within 30 and 90 days. However, it did not significantly improve 2-year survival overall, though stage II patients saw modest survival gains.
(Obeng-Gyasi et al., 2022)	Insurance coverage (as a proxy measure of SES)/ Race, marital status, geographic location	Breast cancer	Surgery	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion was associated with reduced uninsured rates and narrowed disparities in breast conservation therapy and reconstruction among low-income women. However, it did not improve time to surgery, which remained longer than for privately insured patients.
(Ortiz Rueda et al., 2024)	Insurance coverage (as a proxy measure of SES)/ Age, sex, income, education level, geographical location, facility type	Biliary tract cancers (gallbladder carcinoma, extrahepatic cholangiocarcinoma, intrahepatic cholangiocarcinoma)	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion was associated with increased receipt of surgery and multimodal treatment for Stage I–III biliary tract cancer among uninsured/Medicaid patients, leading to improved survival. No significant changes were observed for patients with other insurance types.
Parina et al. (2024)	Insurance coverage (as a proxy measure of SES)/ Age, sex, race, geographical location, income, education	Hepatobiliary cancers	Surgery, Radiotherapy, Systemic therapy (chemotherapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	The impact of Medicaid expansion was mixed. For example, the rate of initiating treatment within 30 days of diagnosis decreased over time in both expansion states and non-expansion states, but on difference-in-differences analysis, was significantly worse in expansion states.
(Primm et al., 2024)	Insurance coverage (as a proxy measure of SES)/ Race, age, sex, facility type, geographical location, income	Breast, cervical, colon, non-small cell lung cancer	Surgery, Radiotherapy, Systemic therapy (chemotherapy, immunotherapy, hormonal therapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion was associated with improved timely treatment (receipt of treatment within 60 days); this impact was the largest among uninsured patients and patients on Medicaid.
Sabik et al. (2022)	Insurance coverage (as a proxy measure of SES)/ Age, race, income, education, employment, geographic location	Breast cancer	Surgery, Radiotherapy	The Massachusetts healthcare reform was a legal reform aimed at providing universal healthcare coverage through public insurance expansions, mandates, and government subsidies. The law set out	Healthcare reform was associated with increased receipt of breast cancer treatment; specifically, increased receipt of mastectomy with reconstruction (a treatment

(*continued on next page*)

Table 5 (*continued*)

Author (Year)	Primary Social Disadvantage/ *Secondary Social Disadvantage*	Cancer type	Treatment Type	Brief Description of Intervention, Service or Model of Care	Brief Description of Impact on Access to Cancer Treatment or Health Equity
				minimum coverage standards, and requirements for the provision of employee health insurance.	historically less available to low-income and uninsured women) while decreasing mastectomy without reconstruction.
(Schpero et al., 2024)	Insurance coverage (as a proxy measure of SES)/ Race	Breast, GI, GU, gynecologic, hematologic, neurologic/brain, skin, thoracic, Other	Therapeutic clinical trials (not otherwise specified)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion was not associated with an increase in access to clinical trials overall, but was associated with increased clinical trial participation in specific states that mandated Medicaid coverage of routine costs of clinical trial participation.
(Shi et al., 2024)	Insurance coverage (as a proxy measure of SES)/ Age, race, geographic location, income, insurance type	Breast cancer	Surgery, Systemic therapy (chemotherapy, targeted therapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion was associated with improved overall receipt of guideline-concordant treatment, and an increase in receipt of treatment within < 60 days.
Takvorian et al. (2020)	Insurance coverage (as a proxy measure of SES)/ Age, sex, race, education, geographic location, income, insurance status	Breast, colon, non-small cell lung cancer	Surgery, Radiotherapy, Systemic therapy (chemotherapy, immunotherapy, hormonal therapy)	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion did not result in changes to the rate of timely cancer-directed treatment, although it did result in a decrease in the rate of people without insurance, and improvements in the rate of early-stage cancer diagnosis.
(Tamirisa et al., 2024)	Insurance coverage (as a proxy measure of SES)/ Race	Breast cancer	Surgery	Medicaid Expansion (ME) under the Affordable Care Act (ACA)	Medicaid expansion reduced delays in breast cancer surgery for racial and ethnic minorities compared to White patients. This difference not seen in people with private insurance. In states that did not expand Medicaid, racial disparities in time from diagnosis and surgery worsened. Overall, study findings suggest that expanding Medicaid may help reduce racial inequalities in timely access to surgical treatment for early-stage breast cancer.

navigation in which nurse navigators were responsible for monitoring system-based feedback (as above) and tracking patient milestones, engaging directly with patients who were experiencing challenges in moving through the treatment trajectory (Charlot et al., 2022; Cykert et al., 2020). Additionally, two studies reported on interventions in which the location of service delivery was changed with the goal of improving access to treatment (Scrace and Margolis, 2009; Tang et al., 2021).

Evaluations of these interventions demonstrated that most were effective in improving access to cancer treatment. For example, Anwuri et al. (2013) report on a multi-component intervention to improve access to therapeutic clinical trials for people experiencing social disadvantage related to racial and ethnic identities that included a) obtaining explicit support and a mandate for change from senior leadership at the institution; b) institutional policy changes to define clinical trial accrual targets; c) creation of institutional processes and infrastructure for data analysis and reporting on clinical trial accrual, and the provision of feedback to clinical trial staff and investigators on accrual. Evaluation demonstrated increased overall therapeutic clinical trial enrollment, and significant increase in the number of 'minorities' participating in clinical trials (Anwuri et al., 2013). Other studies demonstrated that institutional and system-focused interventions improved timeliness of treatment initiation (Charlot et al., 2022; Chen et al., 2023; Tang et al., 2021), improved treatment completion (Cykert et al., 2020), and reduced disparities for equity-seeking groups (Anwuri et al., 2013; Charlot et al., 2022; Chen et al., 2023; Cykert et al., 2020) (see Table 5).

Macro-level: policy-focused interventions

The majority of studies included in our review reported on broader, macro-level policy-focused interventions (n = 32); specifically, almost all of these (n = 31) reported on evaluations of Medicaid expansion policies stemming from the passage of the Affordable Care Act (ACA), an American federal law, passed in 2010, aimed at expanding access to health insurance. Among other provisions, the ACA provided a pathway for states to expand Medicaid (government-provided health insurance for low-income people), thereby increasing access to many forms of cancer treatment by providing insurance coverage for those treatments. One additional study reported on a broad policy reform in Massachusetts which aimed to provide universal health care coverage through mandated minimum insurance coverage, government-subsidized insurance and public insurance expansions (Sabik et al., 2022).

Evaluations of policy-based interventions demonstrate mixed results (see Table 5). Some studies indicate that these interventions improved access to cancer treatment (increased receipt of treatment, improved timeliness of treatment, reduced delays in treatment), whereas other

studies showed no impact on access to cancer treatment. Several studies showed that Medicaid expansion simultaneously improved access to treatment on a population level, while also reducing disparities between equity-seeking groups (i.e., the intervention had the greatest impact on equity-seeking groups). Conversely, some studies demonstrated mixed impacts; for example, Lima et al. (2023b) demonstrated that Medicaid expansion increased access to surgical treatment, but only for non-Hispanic White people, and that disparities in access to treatment remained.

Summary of interventions & impact

Across all levels of interventions, impacts on access to cancer treatment varied. In particular, the ways in which 'access' (as a dynamic process) was operationalized and evaluated varied, including aspects of service availability (e.g., opportunity to access treatment), use (e.g., receipt of treatment), and quality (e.g., guideline concordance, treatment completion); in some studies, improved access to treatment was also evaluated as improved throughput (e.g., timeliness of treatment, reduction of barriers). However, not all interventions that improved access also reduced disparities or inequities. We further plotted the level of intervention by the type of impact it had (Fig. 3).

Discussion

In this review, we aimed to map and synthesize the literature describing interventions, services, and models of care designed to improve access to or receipt of cancer treatment among people who are socially disadvantaged. From the studies included, we identified four main categories of interventions spanning micro to macro level interventions, including education-based interventions, patient navigation interventions, institutional and system-focused interventions, and policy-focused interventions. Barriers most frequently targeted included insurance coverage, financial, and transportation. Policy-focused interventions centered on expanding government-funded health insurance programs including cancer treatments primarily through Medicaid expansion in the U.S.; findings from these included studies were mixed and complex, and may have been impacted by state-specific differences in the implementation of Medicaid expansion, and its evaluation (as noted by Laughlin et al., 2023), as well as differences in cancer type and treatment type. Notably, some studies demonstrated that policy-based interventions were effective in not only improving overall access to cancer treatment but also in reducing disparities in access to treatment between groups (e.g., Albright et al., 2021), while others demonstrated that pre-existing disparities in access to cancer treatment were reduced but persisted (e.g., Janopaul-Naylor et al., 2023) or that access improved for some groups but disparities remained (e.g., Lima et al., 2023b). While publicly funded healthcare systems are central to advancing health equity, these findings suggest that assuming the provision of financial coverage for a healthcare service will improve access and therefore health equity is problematic and potentially harmful. Indeed, several authors noted that insurance coverage must correspond with sufficiently available services, and hypothesized that sudden surges in service demand (corresponding to an influx of insured patients who were previously uninsured) may overwhelm an already stretched oncology system, paradoxically worsening access to treatment by lengthening wait times (Parina et al., 2024; Takvorian et al., 2020). Moreover, many barriers to accessing cancer treatment are not addressed by publicly funded health insurance; thus, while health insurance is foundational, additional interventions are needed to address patient, provider, and health systems barriers.

Patient navigation was initially conceptualized as an equity-promoting intervention in the context of breast cancer screening and diagnosis (Freeman and Rodriguez, 2011) but has since expanded to a range of patient navigation models, some of which include navigation up to the point of cancer treatment initiation or through to completion of treatment. Given that a core function of patient navigation is the elimination of barriers to timely and high-quality care, it continues to hold promise as an intervention with the capacity to promote more equitable and accessible care. In the studies included in our analysis, we noted

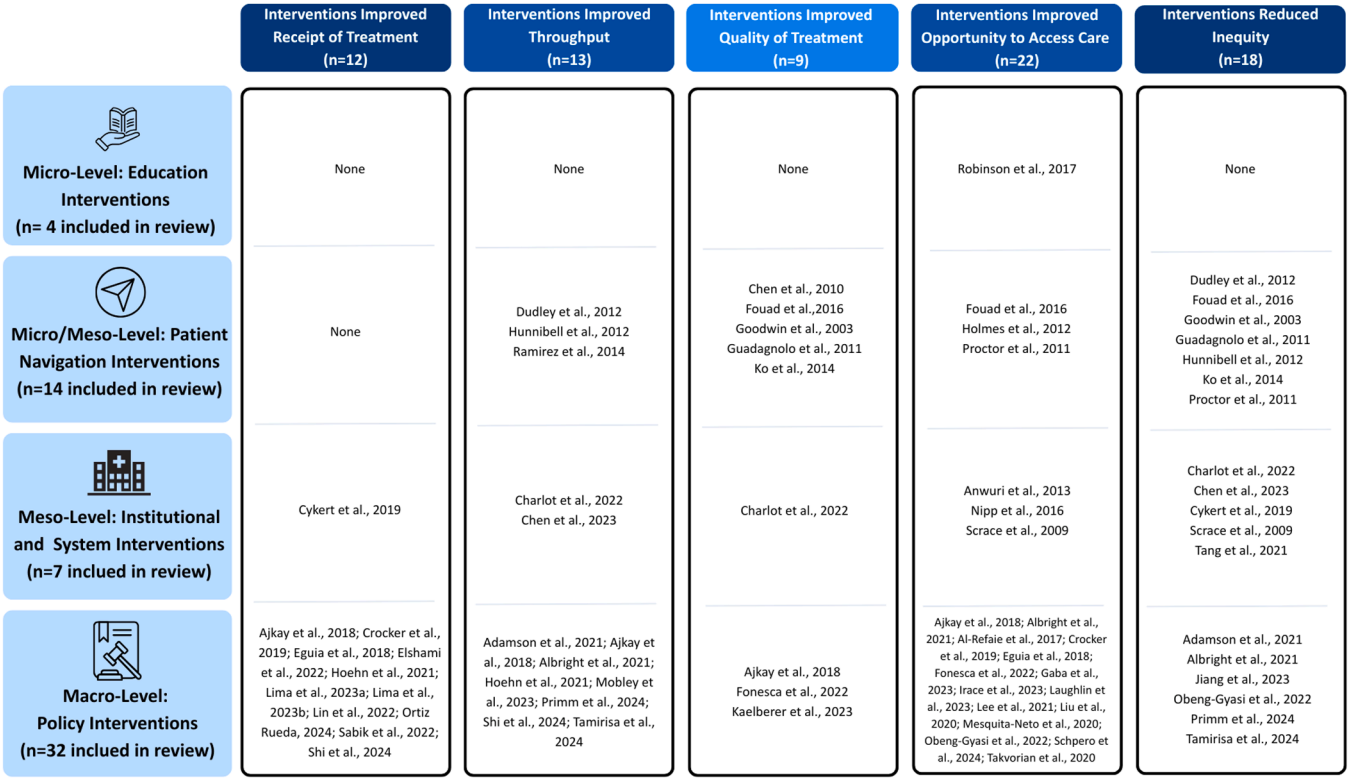


Fig. 3. Intervention Level by Intervention Impact.

variations in how patient navigation was designed and delivered (including variations on who is navigating, where navigation services are housed, the components of patient navigation services, and how long patients are navigated through the cancer continuum), yet navigation interventions were consistently designed to address multiple barriers, often working to address both health system and individual patient barriers in tandem. Notably, several patient navigation interventions were culturally tailored to the specific populations served (Chen et al., 2010; Dockery et al., 2018), and in some cases, there was evidence of community engagement in the design (Guadagnolo et al., 2011) and implementation (Dudley et al., 2012a) of the intervention. Community engagement, partnership, collaboration, and co-design are particularly important in the context of socially disadvantaged populations who have experienced historical and ongoing oppression and disenfranchisement, and have been identified as key pathways to promoting health equity in cancer care (Alcaraz et al., 2020; American Cancer Society, n.d.; Canadian Partnership Against Cancer, N.D.; Patel et al., 2020).

Although patient navigation is a flexible, tailorable and often effective intervention in improving access to cancer treatment and mitigating gaps in care, it also points towards symptoms of dis-integrated and dysfunctional cancer care systems (Thorne and Truant, 2010a), indicating the need for broader health system reform and multi-level intervention. For example, Charlot et al. (2022) and Cykert et al. (2020) reported on two broad-based multi-component interventions that addressed system, institutional, patient and clinician barriers by combining patient navigation with additional interventions. The system-based intervention consisted of: (a) a real-time warning system derived from automated uploads from electronic health records delivering alerts when patients missed a scheduled appointment or did not reach an expected milestone in care (e.g., no surgery or radiotherapy received within 90 days); (b) a nurse navigator who accessed the warning system daily, re-engaged with patients directly to identify and address barriers to care (e.g., logistical, financial, mistrust, cultural, miscommunication), and then engaged with physicians to address treatment milestone issues, and noted resolutions in the warning system; and (c) a physician champion (practicing oncologist) at each center who continually reminded staff of the study, and provided quarterly feedback reports for the cancer center on treatment completion stratified by race. In this example, the power of electronic health records and routinely collected data is leveraged alongside tailored patient navigation, and institutional structures to synergistically address health system gaps. While we did not find published evaluations of patient navigation interventions outside of the U.S., Thorne and Truant's (Thorne and Truant, 2010) critique of patient navigation was developed in the Canadian context of publicly-funded 'universal' healthcare, where evidence suggests that the cancer care system is complex and siloed (Horrill et al., 2025a; Horrill et al., 2022), indicating that patient navigation interventions may be relevant and necessary in contexts outside of the U.S.

Multi-level systems interventions are promising, and the potential to harness existing data through electronic health systems suggests such interventions are scalable, and possibly more sustainable. However, in our own programs of research and interactions with organizational and health systems leaders in cancer care, we are also acutely aware of narratives that frame advancing equity as time-consuming and expensive. Yet this review suggests that smaller-scale and potentially less costly shifts may also be effective in advancing cancer health equity and that interventions at all levels are needed, ranging from those at the patient and provider level to those at the policy level. For example, Chen et al. (2023) pilot tested an appointment scheduling intervention implemented within a tertiary care radiation oncology clinic in which institutional structures and process were developed to offer consultation with an oncologist on the same day as a referral was received. This included the creation of a designated intake team, daily appointment slots held for new patient consultations, and a priority scheduling system to allocate patients to oncologists. The intervention was effective in significantly reducing time from referral to first treatment in all groups, while also decreasing the time from referral to consultation, and improving appointment attendance rates; the impact was most significant in Black and Latino patients (Chen et al., 2023). Notably, in this example, the intervention used existing staff but focused on developing new *processes* and ways of *organizing and delivering care* suggesting that small shifts can lead to significant gains.

The mixed findings observed for some interventions, particularly policy-focused interventions and patient navigation programs, may reflect differences in study design, implementation, and context across studies. Variability in geographical context, health system structures, and implementation approaches may influence outcomes. In addition, studies targeted different populations and used varied measures of treatment access (e.g., treatment initiation, timeliness, or completion), which may contribute to inconsistencies in reported effects. These variations also highlight gaps in understanding which intervention components are most effective across different health care system settings and disadvantaged populations.

Finally, we observed several notable patterns across the included studies. First, most studies were observational in design, with relatively few randomized or pragmatic trials; evaluations rarely included patient, community or healthcare provider perspectives. Second, social disadvantage was often operationalized through proxies such as Medicaid enrollment or income level; however, many studies also considered additional variables, including race/ethnicity, rurality, or education. Despite a rich theoretical knowledge base on the intersectional nature of social disadvantage and health, none of the included studies *explicitly* applied an intersectional framework. Third, while a wide range of intervention settings were represented, nearly all interventions were implemented and evaluated in the U.S. These patterns may be rooted in several interconnected issues.

The dominance of American literature may be related to variations in how social disadvantage is conceptualized in different settings, and a limitation of our search strategy. This may also be reflective of well-known biases in academic publishing with a preference for western/European research, although several other factors may be at play. A long tradition exists within the U.S. of health equity and health justice scholars working to expose and dismantle systems of oppression through theoretically informed critical social science (Lett et al., 2022). However, in line with the growing attention to health equity and social determinants of health that began in the early 2000's, the past decade has seen calls for applications to major American research funding bodies and academic research papers related to health inequities, with a subsequent explosion of epidemiological research on health disparities resulting in what Lett and colleagues (2022) have termed *equity tourism*. The impacts of health equity tourism include a proliferation of research loosely concentrated on health equity that is reactionary, and often conducted by scholars who have not sufficiently or critically engaged with health equity theories, using methodologies that may be ineffective in developing meaningful evidence at best, and/or further perpetuating inequities, systemic exclusion and oppression at worst (Lett et al., 2022). The lack of explicit use of intersectional frameworks may be a by-product of equity tourism, but may also be a result of biomedical journal reporting requirements which limit word count, and tend to emphasize reporting of methodology and findings rather than theoretical and epistemological foundations.

The choice of methodological approach (for example, retrospective observational designs) may be related to equity tourism, while also being influenced by normative hierarchies of evidence. These hierarchies have been driven by the evidence-based medicine movement and related de-valuing of context-specific, smaller scale, community engaged, qualitative and mixed-methods research that may actually hold more promise in developing relevant, effective, community-driven interventions and evaluation mechanisms. The over-reliance on administrative databases to conduct health equity research is evident in the health services research landscape – one that is dominated by

research using proxy measures for social disadvantage, and obscures the complex social and structural forces at play that create and maintain social disadvantage (Petteway, 2023). While this mode of health equity knowledge production may be widely accepted as producing 'strong' evidence, it tends to be reductionist and excludes/silences the communities that the research purports to benefit (Petteway, 2023). As a result, interventions developed from this knowledge base may not be effective in improving equitable access to care or health equity.

Implications & future directions

Our review identified several notable gaps in the literature, highlighting key areas for future research. Most broadly, there was general dearth of research on interventions specifically designed to improve access to cancer treatment for socially disadvantaged populations. More specifically, we noted an absence of the integration of intersectional perspectives on social disadvantage. While studies overwhelmingly focused on income/socioeconomic status (and Medicaid enrollment as a proxy measure) and race as forms of social disadvantage requiring intervention to advance toward equitable access to cancer treatment, integrating an intersectional lens to consider *how* various forms of social disadvantage work together to compound and further intensify systems of oppression would deepen this body of research. Theoretically informed methods that more fully account for the complexities of health inequities are needed to guide research and intervention development. For example, Krieger's Ecosocial Theory accounts for the interplay between social and biological processes across multiple levels and has guided corresponding methodological approaches to population health research (Krieger, 2019). Implementation science frameworks that draw specific attention to health equity, such as the Health Equity Implementation Framework (Woodward et al., 2019) or the Conceptual Framework for Equity-Focused Implementation Research (Eslava-Schmalbach et al., 2019), may also be useful in advancing the evidence base. Critically, the lack of intersectional perspectives may have consequences for our understanding of the impacts of interventions. For example, the reported effectiveness of some interventions may be skewed or incomplete because of this gap; alternatively, interventions and or evaluations targeting only a single dimension of disadvantage may inadvertently neglect or worsen barriers resulting from other dimensions of disadvantage. Similar gaps are noted in implementation science research to promote health equity more broadly, including interventions that insufficiently account underlying structural factors and inadequate methods for evaluation (Fitzpatrick et al., 2023).

In the International Agency for Research on Cancer's report on reducing social inequalities in cancer, Wild notes that for socially disadvantaged people, not only are cancer outcomes and access to care inequitable, but cancer is also *lived* and *experienced* differently (2019, p. 4). As such, the integration of patient-reported outcomes (PROs) and patient-reported experiences (PREs) in the evaluation of interventions is especially important to provide a more robust understanding of the impacts of interventions on patients' status, quality of life, and experiences of care (Casaca et al., 2023). These data also play a pivotal role in understanding patient needs and priorities, and how interventions may need to be modified to best meet these (Casaca et al., 2023; Sharma et al., 2024).

In advance of carefully considering how to integrate PROs and PREs, developing interventions that are meaningful and impactful to socially disadvantaged communities requires that those communities be at the center of these efforts. In addition to engagement and partnership with socially disadvantaged groups to co-develop interventions, we advocate for greater use of mixed-methods and community-based participatory action-based research approaches that are theoretically grounded in social justice aims, including attending to power hierarchies that have historically and continue to shape knowledge production. These approaches support working *with* communities to identify the most pressing barriers to accessing cancer care *and* determine the desired outcomes of eventual interventions. Such approaches prioritize genuine partnership, including equalizing power relationships, invite the integration of community values and experiential knowledge, and ensure that research benefits the communities themselves in addition to academic and clinical partners (Kothari and Smith, 2022; Wallerstein and Duran, 2010). Despite recognition of the imperative of community engaged research and its potential to address health inequities, significant structural and institutional challenges have been identified to meaningful and sustained engagement (Hippalgaonkar et al., 2024), in addition to the de-valuing of this form of knowledge production as discussed above. Yet community-engaged research has been shown to increase intervention effectiveness, result in more contextually appropriate and culturally relevant interventions, and lead to interventions that are more sustainable (Wallerstein and Duran, 2010), lending further support to its importance in advancing cancer health equity.

Finally, the dominance of U.S.-based interventions has important implications. First, it is plausible that the fragmented, insurance-based healthcare system and unique sociopolitical context have influenced the types of interventions developed to address these gaps, and may not be relevant to other healthcare systems. However, other reviews of access to cancer care services have noted that cancer services tend to be siloed, suggesting that this issue also exists within publicly funded healthcare systems (Bourgeois et al., 2024; Horrill et al., 2019). Second, a key principle underpinning implementation science broadly, and the implementation and evaluation of interventions specifically, is the recognition of context as multilevel and crucial (Silver et al., 2023). Interventions are known to be highly sensitive to context, which influences intervention design and effectiveness (Moore et al., 2021). Thus, the interventions included in this review were designed and implemented in a unique sociopolitical context (e.g., insurance-based healthcare) and may not be relevant in other contexts. At the same time, implementation science scholars insist that interventions must always be developed for context – interventions known to be effective in some contexts must be carefully adapted for new contexts (Moore et al., 2021). Implementation science frameworks such as PARIHS propose 'context' as a key determinant of successful intervention implementation (Harvey and Kitson, 2016), while the ADAPT framework provide guidance on adaptation of interventions to local contexts (Moore et al., 2021). Thus, researchers looking to draw on this review to inform their own intervention implementation and evaluation will need to account for their unique social, economic, political and healthcare contexts.

Strengths & limitations

There may be several limitations to our review. First, we limited our search to English-language documents only and may have missed relevant literature. Second, most included studies were U.S.-based, which may limit the generalizability of our findings to other health systems, particularly those that are publicly funded. Few studies were conducted in rural areas, which may limit the conclusions that can be drawn regarding effectiveness of interventions in these settings. Finally, we included several complex concepts (healthcare access, social disadvantage) which may be conceptualized differently in differing contexts and that can be difficult to operationalize into a search strategy, and as a result, we may have missed relevant studies. However, a robust search strategy developed by an academic librarian with significant subject matter expertise and a strong interdisciplinary team with significant content, methodological and clinical expertise contributed to the strength of our review. Additionally, our search was not limited by date or geographical region. Rigor was enhanced by following an established methodology, a clearly described and published protocol, and through the use of multiple reviewers and data extractors. Finally, the relevance of our findings is enhanced through stakeholder engagement, including patients and community members with lived experience of cancer and social disadvantage, and our ongoing collaboration with

interdisciplinary leaders, decision makers, clinicians and researchers within the cancer care system.

Conclusion

Cancer health equity, and in particular, equitable access to cancer treatment, is a pressing concern given the advancements in the detection and treatment of cancer in recent decades alongside persistent inequitable cancer outcomes for socially disadvantaged groups. Alcaraz et al. remind us that "continuing on a path that more rapidly improves cancer outcomes for some segments of the population over others will contribute to persistent (and potentially widening) disparities" (2020, p. 31). Freeman and Rodriguez have dubbed this the discovery-delivery disconnect – the gap between "what we know and what we do for *all* people" (Freeman and Rodriguez, 2011, emphasis added). Thus, a commitment to advancing cancer health equity (and reminding ourselves of the meaning of equity – fair access to resources depending on need) requires evidence-informed strategies that direct resources where they are most needed to facilitate equitable access to high-quality cancer treatment and care. Our review highlighted interventions ranging from patient education interventions to policy-based insurance interventions, with mixed findings of effectiveness in improving access to cancer treatment for socially disadvantage people. While it appears that institutional and system-level interventions may hold promise for improving equitable access to cancer treatment, implementation research driven by community priorities with interventions designed to respond to local contexts should be prioritized to ensure relevancy, improve effectiveness, and enhance reciprocity.

CRediT authorship contribution statement

Tara Horrill: Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Formal analysis, Writing – original draft. **Amber Bourgeois:** Conceptualization, Funding acquisition, Data curation, Writing – review & editing. **Noor Breik:** Data curation, Validation, Writing – review & editing. **Max Kleijberg:** Conceptualization, Funding acquisition, Writing – review & editing. **Janice Linton:** Conceptualization, Funding acquisition, Methodology, Writing – review & editing. **Morgan Stirling:** Data curation, Validation, Writing – review & editing. **Kelli Stajduhar:** Conceptualization, Funding acquisition, Writing – review & editing.

Ethical approval & consent to participate

Ethical approval was not required. No human participants.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ssmhs.2026.100238.

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